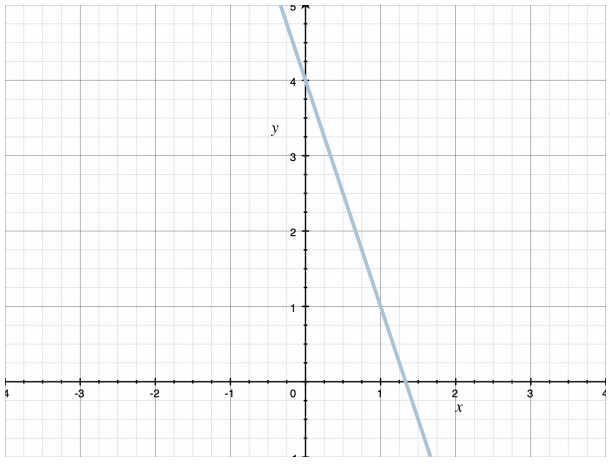
Algebra 1
Final Exam Solutions

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MATH

Algebra 1 Final Exam Answer Key

1. (5 pts) A B C D **E**
2. (5 pts) A B C **D** E
3. (5 pts) A **B** C D E
4. (5 pts) **A** B C D E
5. (5 pts) A B **C** D E
6. (5 pts) A B C **D** E
7. (5 pts) A B **C** D E
8. (5 pts) **A** B C D E
9. (15 pts)



10. (15 pts) $x = \dfrac{-5 \pm \sqrt{73}}{4}$
11. (15 pts) Domain: $x \geq 2$, Range: $y \geq 0$
12. (15 pts) Odd

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Algebra 1 Final Exam Solutions

1. E. The Distributive Property removes the parentheses from the expression. To distribute, multiply the factor outside the parentheses by each term inside the parentheses.

2. D. Plug 2 in for a, 1 in for b, and 4 in for c using parentheses.

$$b^0 + 3(5a - 2b + c^2) - 4a \div 2$$

$$(1)^0 + 3(5(2) - 2(1) + (4)^2) - 4(2) \div 2$$

Simplify using PEMDAS (remember that anything raised to the power of 0 is 1).

$$1 + 3(10 - 2 + 16) - 8 \div 2$$

Simplify the expression inside the parentheses and the division.

$$1 + 3(8 + 16) - 8 \div 2$$

$$1 + 3(24) - 4$$

Simplify the multiplication and then add and subtract from left to right.

$$1 + 72 - 4$$

$$73 - 4$$

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69

3. B. Distribute the -1 across the second set of parentheses and remove the first set of parentheses.

$$(8x^2 - 3x) - (6x^2 - 9x + 4)$$

$$8x^2 - 3x - 6x^2 + 9x - 4$$

Combine like terms.

$$2x^2 + 6x - 4$$

4. A. Let n be the first number and $n + 1$ be the second consecutive number. Solve for n.

$$n(n + 1) = 132$$

Distribute.

$$n^2 + n = 132$$

$$n^2 + n - 132 = 0$$

$$(n + 12)(n - 11) = 0$$

$n + 12 = 0$ or $n - 11 = 0$

$n = -12$ or $n = 11$

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Since 132 is positive, then the two consecutive numbers must be positive, since the product is positive. Therefore, $n = 11$ and $n + 1 = 12$. The consecutive numbers are 11 and 12.

5. C. Use polynomial long division to simplify.

$$
\begin{array}{r}
x^2 + 4x + 8 + \frac{24}{x-2} \\
x-2 \overline{)\, x^3 + 2x^2 + 0x + 8} \\
-(x^3 - 2x^2) \\
\hline
4x^2 + 0x \\
-(4x^2 - 8x) \\
\hline
8x + 8 \\
-(8x - 16) \\
\hline
24
\end{array}
$$

6. D. Write out all the factors of each term.

$$3a^5b^3 - 18a^3b^3 - 21a^2b^4$$

$$3 \cdot a \cdot a \cdot a \cdot a \cdot a \cdot b \cdot b \cdot b - 2 \cdot 3 \cdot 3 \cdot a \cdot a \cdot a \cdot b \cdot b \cdot b$$

$$-21 \cdot a \cdot a \cdot b \cdot b \cdot b \cdot b$$

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The only factors that are shared by all three terms are a 3, an $a \cdot a$, and a $b \cdot b \cdot b$, so the greatest common factor is $3a^2b^3$. When we factor out the $3a^2b^3$, we have to divide each term by $3a^2b^3$.

$$3a^2b^3(a^3 - 6a - 7b)$$

7. C. Subtract 6 from both sides of the inequality.

$$6 - 2x \leq 14$$

$$6 - 6 - 2x \leq 14 - 6$$

$$-2x \leq 8$$

Divide both sides by -2 and, since we're dividing by a negative, remember to switch the direction of the inequality from $\leq$ to $\geq$.

$$\frac{-2x}{-2} \geq \frac{8}{-2}$$

$$x \geq -4$$

8. A. Interpret each part of the statement starting with "the sum of 5 and a number."

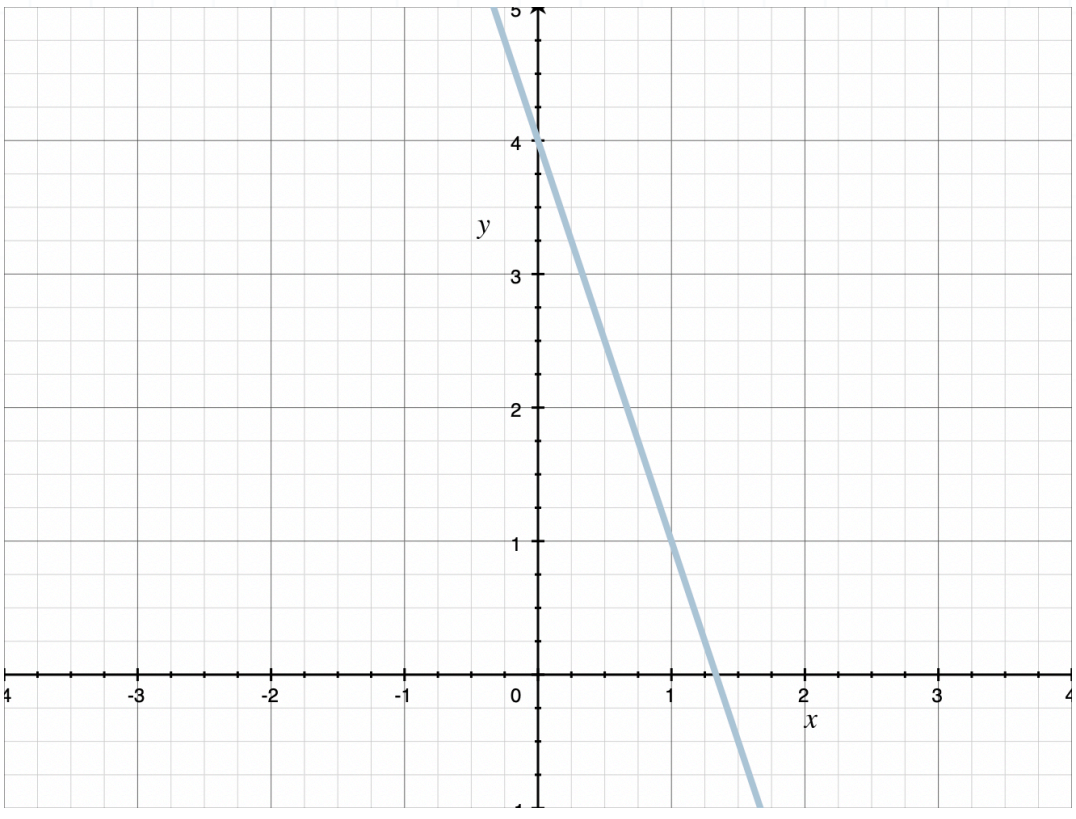
$$5 + x \text{ or } x + 5$$

Next, "product" means to multiply.

$$10(x + 5)$$

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9. Draw the x- and y-axes and make sure to label tick marks and axes. The y-intercept for $y = -3x + 4$ is 4, so put a point on the y-axis at 4. The slope is $-3/1$, which means from the y-intercept we need to go down 3 units and to the right 1 unit to find the next point on the graph. Find one more point using the slope, then draw a straight line that goes through the points and extends past them in both directions.



10. Use the quadratic formula to find the solutions. In this problem with $2x^2 + 5x - 6 = 0$, we identify $a = 2$, $b = 5$, and $c = -6$.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

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$$x = \frac{-5 \pm \sqrt{5^2 - 4(2)(-6)}}{2(2)}$$

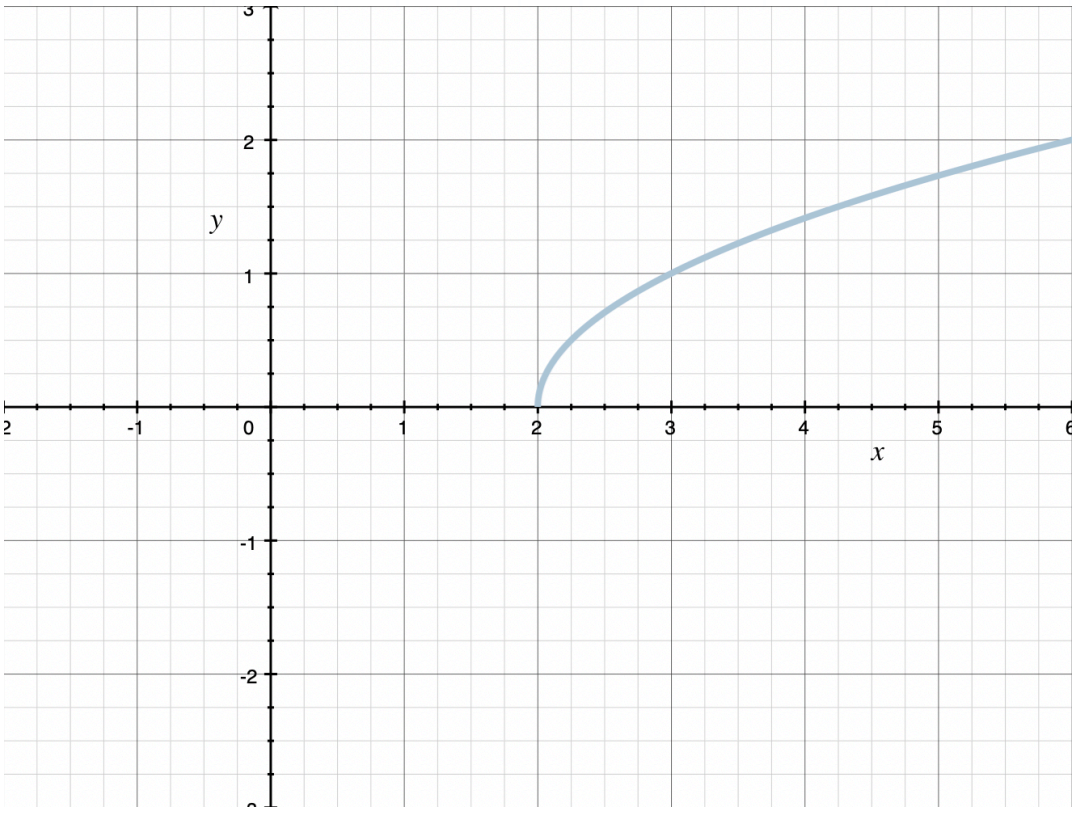
$$x = \frac{-5 \pm \sqrt{25 + 48}}{4}$$

$$x = \frac{-5 \pm \sqrt{73}}{4}$$

Then the solutions are

$$x = \frac{-5 - \sqrt{73}}{4} \text{ and } x = \frac{-5 + \sqrt{73}}{4}$$

11. Graph the function $f(x) = \sqrt{x - 2}$.



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The x-coordinate of the leftmost point is 2 and the graph continues infinitely to the right. Therefore, the domain is $x \geq 2$. The y -coordinate of the lowest point is 0 and the graph continues infinitely upward. Therefore, the range is $y \geq 0$.

12. To solve algebraically, we need to find the expression for $f(-x)$, so we'll replace every x (in the expression for $f(x)$) with $-x$.

$$f(-x) = (-x)^3 - 3(-x)$$

Remember that

$$(-x)^3 = (-1x)^3 = (-1)^3x^3$$

and

$$(-x) = (-1)x$$

Raising -1 to an odd power gives -1, so

$$f(-x) = (-1)x^3 - 3(-1)x$$

Factor out a -1, and then simplify.

$$f(-x) = -1(x^3 - 3x)$$

Since $f(-x) = -f(x)$, the function is odd. We can see that the graph is symmetric with respect to the origin.

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